Day 2 — Understanding GPT and Large Language Models (LLMs)

Course

GPT + AI Agents + RAG + n8n + API Integration + Business Automation

Lecture Duration

60 Minutes

Level

Beginner

Day 2 Main Goal

Day 1 introduced Generative AI and the overall AI automation ecosystem.

Day 2 goes one level deeper.

Students will learn **what an LLM actually does, how GPT processes information, why prompts matter, why models sometimes give different answers, and how to think about an LLM when designing business systems.**

By the end of Day 2, students should understand:

- What a Large Language Model is
- What GPT means
- Model vs application vs agent
- What parameters are
- What tokens are
- Tokenization
- Embeddings at a conceptual level
- Context and context windows
- How next-token prediction works
- What "attention" means conceptually
- Why transformers matter
- How a response is generated
- Deterministic vs variable outputs
- Temperature and randomness
- Input and output tokens
- Instructions vs user data

- Model knowledge vs supplied context
- Why LLMs can appear to reason
- Why LLMs make mistakes
- How to choose tasks appropriate for an LLM
- How LLMs fit into AI business automation

---

1. Day 2 — 60-Minute Schedule

Time	Topic
0–5 min	Day 1 review
5–12 min	What exactly is an LLM?
12–18 min	GPT and transformer concepts
18–27 min	Tokens and tokenization
27–34 min	Context and attention
34–42 min	How GPT generates responses
42–48 min	Temperature, variability, and output control
48–53 min	LLM strengths and weaknesses
53–57 min	Business examples and live demonstration
57–60 min	Quiz, recap, and homework

---

2. Opening Review — 0 to 5 Minutes

Start by asking students five questions from Day 1.

Question 1

What does GPT stand for?

Answer:

Generative Pre-trained Transformer.

Question 2

What does LLM stand for?

Answer:

Large Language Model.

Question 3

What is a prompt?

Answer:

The instructions and/or information provided to an AI model.

Question 4

What is hallucination?

Answer:

When an AI generates information that appears plausible but is unsupported or incorrect.

Question 5

Does GPT automatically know a company's live inventory?

Answer:

No.

It would normally require access to:

- An API
- Database
- Business application
- Tool
- Or data explicitly provided in context

---

3. Introduce Today's Question

Tell students:

> Yesterday we learned what GPT can do.
>
> Today we will answer a deeper question:

> **What happens between the moment we send a prompt and the moment GPT produces an answer?**

Use this diagram:

```
Prompt
   ↓
???
   ↓
AI Response
```

Today's lecture explains the middle.

---

4. What Exactly Is an LLM? — 5 to 12 Minutes

LLM = Large Language Model

Break the name into three pieces.

Large

"Large" generally refers to the scale of:

- Training data
- Model parameters
- Computation
- Model architecture

It does **not** simply mean:

> It has a large database of answers.

That is a common misconception.

---

5. What Is a Model?

A model is a mathematical system that has learned patterns from training data.

Very simplified:

```
Training Data
     ↓
Learning Process
     ↓
Model Parameters
     ↓
Trained Model
```

Later:

```
New Input
   ↓
Trained Model
   ↓
Output
```

---

6. What Are Parameters?

A parameter is a learned numerical value inside a neural network.

Students do not need the mathematics.

Use this analogy:

Imagine millions or billions of tiny adjustable settings.

During training, those settings are adjusted so that the model becomes better at predicting language.

Conceptually:

```
Before Training

Model Settings
[? ? ? ? ? ? ?]

        ↓

Training

        ↓

```

```
Learned Settings
[✓ ✓ ✓ ✓ ✓ ✓ ✓]
```

Important:

Parameters do **not** work like rows in a normal database.

You cannot normally say:

> Parameter 5,481 stores the answer to "What is Python?"

Knowledge is distributed across the network.

---

7. What Does "Language Model" Mean?

A language model learns patterns in language.

For example:

If you see:

> Please find the invoice attached to this...

You may predict:

> email

or:

> message

An LLM does something conceptually similar, but at enormous scale.

It estimates which token should come next.

---

8. Core Concept

Write this on the screen:

> **An LLM predicts tokens based on previous context.**

That simple process can produce:

- Answers
- Summaries
- Code
- Analysis
- Translations
- Stories
- Structured JSON
- Plans
- Explanations

---

9. Is an LLM Just Autocomplete?

You may hear:

> "GPT is just autocomplete."

That statement is too simplistic.

Autocomplete is useful as an analogy because GPT predicts subsequent tokens.

But modern LLMs learn extremely complex relationships involving:

- Grammar
- Meaning
- Facts
- Programming patterns
- Document structures
- Relationships between concepts
- Multi-step patterns
- Styles
- Instructions

So the better beginner statement is:

> GPT uses next-token prediction, but at a scale and sophistication that enables many complex capabilities.

---

10. GPT Explained — 12 to 18 Minutes

GPT stands for:

G — Generative

It creates output.

Examples:

- Text
- Code
- JSON
- Summaries
- Plans

---

11. P — Pre-trained

Before a user interacts with the model, it has already undergone extensive training.

Simplified:

```
Large Training Corpus
       ↓
Pre-training
       ↓
General-Purpose Model
```

The user does not normally train the whole model every time they ask a question.

---

12. T — Transformer

Transformer is the underlying neural-network architecture associated with GPT.

Avoid mathematics on Day 2.

Explain the main idea:

> Transformers are designed to process relationships between pieces of information in a sequence efficiently.

For language:

```
The customer requested a refund because
the product arrived damaged.
```

The model needs to understand relationships such as:

```
customer → requested
refund → because of damage
product → arrived damaged
```

---

13. Why Transformers Were Important

Older language architectures often processed sequences more rigidly.

Transformers introduced a powerful mechanism called:

Attention

Attention helps the model determine which parts of the context are relevant to other parts.

---

14. Attention — Simple Example

Sentence:

> Sarah gave Emma the laptop because she had finished using it.

Question:

> Who does "she" refer to?

To answer, the model must consider relationships among different words.

Conceptually:

```
she
 ↓
Sarah?
```

```
Emma?
Laptop?
```

Attention mechanisms help the model represent these relationships.

Do not claim that attention equals human understanding.

Say:

> Attention is a mathematical mechanism that helps the model weigh relationships among tokens.

---

15. Another Business Example

Customer says:

> I bought the premium subscription last week, but it still says my account is on the free plan.

Important pieces are:

```
premium subscription
last week
account
free plan
```

A useful model must connect these concepts.

---

16. Tokens and Tokenization — 18 to 27 Minutes

What Is a Token?

A token is a unit of input or output processed by the model.

A token may be:

- A word
- Part of a word
- Punctuation
- A number fragment
- Special symbols

17. Words Are Not Always Tokens

For teaching, show:

```
The customer requested a refund.
```

A beginner may imagine:

```
The
customer
requested
a
refund
.
```

But the actual tokenizer may divide things differently.

For example, a longer unusual word could be broken into pieces.

Conceptually:

```
automation
```

might be handled as one or several token units depending on the tokenizer.

Important:

> A token is not exactly the same thing as a word.

---

18. Why Tokenization Exists

Computers do not directly understand written words.

Language must first be represented numerically.

Simplified:

```
Text
 ↓
Tokenizer
 ↓
Tokens
 ↓
Token IDs
 ↓
Model
```

For example:

```
"Hello world"
```

could conceptually become:

```
[15496, 995]
```

The exact numbers depend on the tokenizer.

---

19. Input Tokens vs Output Tokens

Explain:

Input tokens

Everything sent into the model.

Examples:

- Instructions
- User question
- Conversation history
- Retrieved documents
- Tool results

Output tokens

Everything generated by the model.

Conceptually:

```
Input Tokens
     ↓
    Model
     ↓
Output Tokens
```

---

20. Business Example of Token Usage

Imagine an AI support request.

Input contains:

```
System Instructions      500 tokens
Customer History        1000 tokens
Company Policy          2000 tokens
Customer Question        100 tokens
```

Total input:

```
3600 tokens
```

Model generates:

```
300 output tokens
```

Approximate total processing:

```
3900 tokens
```

Explain:

This matters because token usage affects:

- Context limits
- Speed
- Cost
- How much information can be supplied

---

21. Why Bigger Prompts Are Not Always Better

Students may assume:

> More information always gives better answers.

Not necessarily.

Consider:

```
Customer question
+
500 pages of unrelated company documents
```

This can introduce:

- Noise
- Irrelevant information
- Higher cost
- Slower response
- Retrieval problems

Later, RAG will solve this by selecting only relevant information.

---

22. Context and Context Window — 27 to 34 Minutes

What Is Context?

Context is the information available to the model during generation.

Conceptually:

```
Context
│
├── System instructions
├── User message
├── Conversation history
├── Documents
```

```
├── Tool outputs
└── Retrieved knowledge
```

---

23. Example

Conversation:

```
User:
Our company is Acme Furniture.

Assistant:
Understood.

User:
Write a slogan for our company.
```

The latest question does not repeat:

> Acme Furniture

But that information may still be available from conversation context.

---

24. What Is a Context Window?

The context window is the maximum amount of information a model can consider for a particular interaction.

Think of it as a workspace.

```
┌──────────────────────────────┐
│        CONTEXT WINDOW        │
│                              │
│ Instructions                 │
│ Conversation History         │
│ Retrieved Documents          │
│ Tool Results                 │
│ User Question                │
└──────────────────────────────┘
```

If the workspace becomes too full, information must be:

- Removed
- Summarized
- Retrieved selectively
- Managed differently

---

25. Context Is Not Permanent Memory

This distinction becomes important for agents.

Context

Information currently available to the model.

Long-term memory

Information stored somewhere and retrieved later.

Example:

```
Conversation Context
        ↓
      Model
```

versus:

```
Database
   ↓
Retrieve Memory
   ↓
Context
   ↓
Model
```

The model itself does not necessarily have permanent access to every previous conversation.

---

26. Context vs Training Knowledge

There are two different sources of information.

Learned during training

Example:

```
General knowledge about Python
```

Supplied at runtime

Example:

```
Your company's private refund policy
```

A useful mental model:

```
Model Knowledge
      +
Current Context
      =
Information available for response generation
```

---

27. Why RAG Will Matter Later

Imagine the user asks:

> What is our company's international refund policy?

The model might not know.

Instead:

```
User Question
      ↓
Search Company Documents
      ↓
Retrieve Relevant Policy
      ↓
Place Policy in Context
      ↓
GPT
```

```
       ↓
Grounded Answer
```

That is the basic idea behind RAG.

---

28. How GPT Generates a Response — 34 to 42 Minutes

Now explain the complete generation loop.

Input:

> The customer asked for a...

Potential next tokens:

```
refund        45%
replacement   25%
discount      10%
response       8%
callback       5%
other          7%
```

The model chooses a token based on its probability distribution and generation settings.

Then the process repeats.

---

29. Step-by-Step Generation

Imagine the model generates:

```
The customer requested a refund.
```

Conceptually:

```
The
 ↓
The customer
```

```
  ↓
The customer requested
  ↓
The customer requested a
  ↓
The customer requested a refund
  ↓
The customer requested a refund.
```

At every step:

> What token should come next given everything currently available?

---

30. Important Point

The model does not normally generate the entire response in one magical instant.

It builds the output token by token.

---

31. Why GPT Can Generate Long Logical Answers

Ask:

> If GPT only predicts the next token, how can it write an entire report?

Because each newly generated token becomes part of the context for the next prediction.

Conceptually:

```
Prompt
  ↓
Token 1
  ↓
Prompt + Token 1
  ↓
Token 2
  ↓
Prompt + Token 1 + Token 2
  ↓
Token 3
```

```
 ↓
...
```

This allows coherent sequences to emerge.

---

32. What Happens Internally — Beginner-Level View

Use this simplified pipeline:

```
User Text
    ↓
Tokenization
    ↓
Token Representations
    ↓
Transformer Layers
    ↓
Attention + Learned Patterns
    ↓
Next-Token Probabilities
    ↓
Token Selection
    ↓
Repeat
    ↓
Final Response
```

Tell students:

> This is intentionally simplified. You do not need matrix mathematics to build useful AI applications.

---

33. Embeddings — First Introduction

Students will study embeddings deeply during RAG week.

For Day 2, only introduce the concept.

The model represents tokens and concepts using numerical vectors.

Simple idea:

```
King
Queen
Dog
Cat
Refund
Return
Invoice
Payment
```

Words or concepts with related meanings can have related numerical representations.

This helps models process semantic relationships.

---

34. Semantic Similarity Example

Consider:

> I want my money back.

and:

> Please refund my payment.

These sentences use different words but have similar meaning.

AI systems can recognize that semantic similarity.

Later, RAG systems use embedding-based search for this purpose.

---

35. Randomness and Temperature — 42 to 48 Minutes

Students may ask:

> Why does ChatGPT sometimes answer differently when I ask the same question twice?

Because generation can contain controlled randomness.

---

36. Deterministic vs Creative Output

Suppose the model sees:

> Write a slogan for a coffee shop.

Possible outputs:

```
Wake Up to Better Coffee.
```

or:

```
Every Cup Starts Something.
```

or:

```
Your Daily Moment, Brewed.
```

There may be many valid answers.

---

37. What Is Temperature?

Temperature is a generation control that influences how adventurous or conservative token selection is in systems that expose it.

Simple conceptual explanation:

Lower temperature

More predictable.

```
Focused
Consistent
Conservative
```

Higher temperature

More varied.

```
Creative
Diverse
Less predictable
```

Do not tell students that temperature guarantees quality.

---

38. Business Examples

Classification

Task:

> Is this email Sales, Support, or Billing?

Desired behavior:

```
Consistent
Reliable
Structured
```

We generally prefer lower variability.

Brainstorming

Task:

> Generate 20 unusual marketing campaign ideas.

Desired behavior:

```
Creative
Diverse
Exploratory
```

More variability may be useful.

---

39. Important Modern Engineering Point

Tell students:

Not every model or API exposes exactly the same generation controls.

Therefore, the professional approach is:

> Design around the capabilities of the model/API you are using rather than assuming every LLM supports identical parameters.

---

40. Why Same Prompt Can Produce Different Results

Several reasons:

- Probabilistic token selection
- Different model versions
- Changed context
- Different system instructions
- Updated application behavior
- Tool results
- External information
- Different conversation history

---

41. Prompt + Context Determines Behavior

Use this equation:

```
Model
+
Instructions
+
Context
+
Tools
+
Generation Settings
```

```
=
Observed AI Behavior
```

This equation is much more useful than saying:

> GPT is smart.

---

42. Instructions and User Data

Students should learn that different parts of a prompt serve different purposes.

Example:

```
INSTRUCTION:
Classify the customer message.

CUSTOMER MESSAGE:
"I want a refund immediately."

OUTPUT FORMAT:
JSON
```

The customer message is **data**.

The classification command is **instruction**.

---

43. Why Separating Instructions and Data Matters

Poor prompt:

```
Here is an email, analyze it and maybe tell me what type it is and what to do:
...
```

Better:

```
Role:
Customer Support Classifier

Task:
```

```
Analyze the supplied customer message.

Return:
- category
- sentiment
- priority
- action

Customer Message:
...
```

This concept prepares students for Day 3: Prompt Engineering.

---

44. LLM Strengths — 48 to 53 Minutes

Explain tasks that align well with language models.

1. Summarization

```
Long Report
   ↓
LLM
   ↓
Short Summary
```

---

2. Classification

```
Customer Email
      ↓
     LLM
      ↓
Sales / Billing / Support
```

---

3. Information Extraction

```
"We need 20 laptops delivered to Miami by Friday."
```

Output:

```
{
  "product": "laptops",
  "quantity": 20,
  "location": "Miami",
  "deadline": "Friday"
}
```

---

4. Rewriting

Input:

```
hey your payment failed fix it
```

Output:

> Hello, we noticed that your recent payment was unsuccessful. Please update your payment information when convenient.

---

5. Translation

Transform language A into language B.

---

6. Code Assistance

LLMs can:

- Generate code
- Explain code
- Debug code
- Convert between languages
- Create tests

But generated code must still be reviewed and tested.

---

7. Question Answering

Especially when relevant knowledge is supplied.

---

8. Data Transformation

Input:

Unstructured language.

Output:

Structured JSON.

This becomes extremely useful in automation.

---

45. LLM Weaknesses

Weakness 1 — Hallucination

It may generate unsupported information.

Weakness 2 — Live Information

Without tools, it may not have access to current data.

Weakness 3 — Private Business Information

Without context or RAG, it does not automatically know company-specific facts.

Weakness 4 — Exact Calculation

Language models are not always the best tool for precise arithmetic.

Professional approach:

```
LLM
 ↓
Calculator Tool
 ↓
Exact Answer
```

Weakness 5 — Guaranteed Business Rules

Suppose the rule is:

```
Never approve refunds over $1000 automatically.
```

Do not rely purely on an LLM remembering this in every circumstance.

Enforce important business logic in code or workflow rules.

Example:

```python
if refund_amount > 1000:
    require_human_approval = True
```

---

46. Important Architecture Principle

Teach:

> **Use AI for fuzzy language tasks.**
>
> **Use traditional code for strict deterministic rules.**

Examples:

AI

```
Determine customer sentiment.
```

Code

```
If amount > $1,000 → require manager approval.
```

AI

```
Summarize the complaint.
```

Code

```
Send the ticket to department ID 42.
```

---

47. Hybrid AI System

Professional systems combine both.

```
Customer Message
       ↓
      GPT
       ↓
Intent = Refund
Sentiment = Negative
       ↓
Traditional Business Rule
       ↓
Refund > $1000?
   ┌───────┴───────┐
  Yes              No
   ↓                ↓
Human Review    Continue
```

This is a very important concept for future business automation.

---

48. LLM vs Search Engine

Students often confuse these.

Search Engine

Typically retrieves existing information.

```
Query
  ↓
Search
  ↓
Web Pages
```

LLM

Generates a response.

```
Prompt
  ↓
Model
  ↓
Generated Text
```

---

49. Search + LLM

Modern systems can combine both.

```
Question
    ↓
Search Tool
    ↓
Current Information
    ↓
LLM
    ↓
Synthesized Answer
```

This is more powerful than either component alone.

---

50. LLM vs Database

A database stores explicit records.

Example:

```
Order #93422
Customer: David
Status: Shipped
Amount: $284
```

An LLM should not replace this database.

Instead:

```
Customer:
"Where is my latest order?"
        ↓
       LLM
        ↓
Understand intent
        ↓
Database/API
        ↓
Actual Order Record
        ↓
LLM
        ↓
Friendly Answer
```

---

51. Important Student Principle

Write prominently:

> **LLM = Language Intelligence**
>
> **Database = Stored Facts**
>
> **API = Connection**
>
> **Agent = Decision + Tool Use**

n8n = Workflow Orchestration

RAG = Relevant Knowledge Retrieval

Students will revisit this throughout the course.

---

52. Business Demo — 53 to 57 Minutes

Use this example customer email:

```
Hi,

I've been waiting for my order for 12 days.
Your website said delivery would take 3–5 days.

This is the second time I've contacted support.
I need the laptop for my work.

Order number is LTP-38492.

Please tell me when it will arrive.

Thanks,
Michael
```

---

53. Demo 1 — Ask for Free-Form Analysis

Prompt:

```
Analyze this customer email.
```

Discuss what the model produces.

Ask:

> Is this output easy for another software system to process?

Probably not consistently.

---

54. Demo 2 — Ask for Structured Information

Prompt:

```
Analyze the following customer email.

Return:

Customer Name:
Order Number:
Issue:
Sentiment:
Urgency:
Summary:
Required Action:
```

Possible output:

```
Customer Name: Michael
Order Number: LTP-38492
Issue: Delayed delivery
Sentiment: Negative
Urgency: High
Summary: Customer has waited 12 days for a laptop advertised as arriving within
3–5 days and has contacted support twice.
Required Action: Check current shipping status.
```

---

55. Demo 3 — JSON Output

Prompt:

```
Return the result as JSON.
```

Possible output:

```
{
  "customer_name": "Michael",
  "order_number": "LTP-38492",
  "issue": "shipping_delay",
  "sentiment": "negative",
```

```
  "urgency": "high",
  "requires_order_lookup": true
}
```

Ask:

> Why is this useful?

Expected answers:

- n8n can process it
- Code can process it
- Database can store it
- Conditions can use it
- APIs can use it

---

56. Demo 4 — Expose Missing Information

Ask:

> When will Michael's laptop arrive?

The correct system should recognize:

We do not have the actual shipping data.

We need:

```
Order Number
     ↓
Shipping API
     ↓
Current Tracking Status
```

This demonstrates the boundary between:

language understanding

and:

factual system access.

---

57. Full Future Architecture

Show students:

```
Customer Email
      ↓
     n8n
      ↓
     GPT
      ↓
Extract:
- Customer
- Order ID
- Intent
- Sentiment
      ↓
AI Agent
      ↓
Order API
      ↓
Actual Tracking Information
      ↓
RAG
      ↓
Shipping / Refund Policies
      ↓
GPT
      ↓
Generate Accurate Response
      ↓
Send Email
```

Tell students:

> Everything we learn during the next 28 days builds toward systems like this.

---

58. Five-Minute Student Exercise

Give students:

```
Hello,
```

```
We signed up for the Business Plan yesterday,
but five members of our team still cannot access
the analytics dashboard.

Our account email is admin@samplecompany.com.

We have an important presentation tomorrow morning
and need the dashboard working today.

Regards,
Susan
```

Students should ask an LLM to extract:

```
Customer:
Account:
Problem:
Plan:
Urgency:
Sentiment:
Deadline:
Suggested Next Step:
Missing Information:
```

Expected conceptual output:

```
{
  "customer": "Susan",
  "account": "admin@samplecompany.com",
  "problem": "Five team members cannot access analytics dashboard",
  "plan": "Business Plan",
  "urgency": "high",
  "sentiment": "concerned",
  "deadline": "today",
  "suggested_next_step": "Check account and team-member access status",
  "missing_information": [
    "Affected team member accounts",
    "Current account permissions",
    "System status"
  ]
}
```

---

59. Discussion Question

Ask:

> Which fields came directly from the email?

Examples:

- Customer
- Account
- Problem
- Plan

Ask:

> Which field required interpretation?

Examples:

- Urgency
- Sentiment

Ask:

> Which information would require an external system?

Examples:

- Account status
- Permissions
- Service status

This exercise teaches three information categories:

```
Explicit Data
+
AI Interpretation
+
External Data
```

---

60. Three Types of Information in AI Systems

This is worth emphasizing.

Type 1 — Explicit information

User gives it directly.

Example:

> My order number is 12345.

Type 2 — Inferred information

AI derives it.

Example:

> I have contacted you four times and nobody responded.

AI may classify:

```
Urgency = High
Sentiment = Negative
```

Type 3 — External information

Requires another source.

Example:

```
Current shipping status
```

Needs:

```
Order API
```

Professional AI architects must know the difference.

---

61. Common Beginner Misconceptions

Misconception 1

"GPT searches the internet every time."

Not necessarily.

A model can generate from its available context and learned patterns. Internet access requires a search/browser tool or application capability.

Misconception 2

"GPT stores all facts like a database."

No.

Use databases for exact stored records.

Misconception 3

"If GPT sounds confident, it is correct."

No.

Fluency and correctness are different.

Misconception 4

"Longer prompts are always better."

No.

Relevant context is more important than unnecessary volume.

Misconception 5

"AI should replace all business rules."

No.

Strict rules often belong in deterministic code.

Misconception 6

"One AI model should do everything."

Often, professional systems combine:

```
LLM
+
Database
+
Search
+
APIs
+
Code
+
Workflow Automation
+
Human Review
```

---

62. Day 2 Key Architecture

Show:

```
                  USER
                    ↓
               APPLICATION
                    ↓
                   LLM
             ┌──────┴─────┐
             ↓            ↓
     Learned Knowledge   Context
                          │
                    ┌─────┼─────┐
                    ↓     ↓     ↓
                   RAG   APIs  Memory
                    │     │
```

```
                ↓      ↓
             Documents Live Data
```

Explain:

The LLM is central, but it is **not the entire system**.

---

63. Day 2 Quiz

Question 1

What is the primary basic operation of an LLM?

A. Searching Google
B. Predicting tokens based on context
C. Reading databases automatically
D. Executing programs directly

Answer: B

Question 2

What does the "P" in GPT mean?

A. Predictive
B. Prompt
C. Pre-trained
D. Programmed

Answer: C

Question 3

What is a token?

A. Always exactly one word
B. A unit processed by the language model
C. A database row
D. An API

Answer: B

Question 4

What is tokenization?

A. Training a new AI model
B. Converting text into token units
C. Searching a vector database
D. Calling an API

Answer: B

Question 5

What is context?

A. Information available to the model during generation
B. Only information learned during training
C. Only the current user question
D. A type of database

Answer: A

Question 6

What is a context window?

A. A browser window
B. The amount of information a model can work with during an interaction
C. An AI training program
D. An API endpoint

Answer: B

Question 7

Which system should normally store exact order status?

A. LLM
B. Database / order management system

C. Prompt
D. Temperature

Answer: B

Question 8

Which task is particularly suitable for an LLM?

A. Determining sentiment from a customer email
B. Guaranteeing exact financial ledger balances without tools
C. Physically shipping a package
D. Replacing every database

Answer: A

Question 9

What does attention help a transformer do?

A. Turn on the computer
B. Represent relationships among tokens in context
C. Access all websites automatically
D. Store customer records permanently

Answer: B

Question 10

Which is generally a better architecture?

A.

```
GPT → Guess current order status
```

B.

```
GPT → Order API → Actual order status → GPT response
```

Answer: B

64. True or False Round

1. Every token is exactly one English word.

False

2. LLM responses may vary between runs.

True

3. Context and training knowledge are exactly the same thing.

False

4. An LLM can be useful for classification.

True

5. A database is generally better than an LLM for storing exact customer balances.

True

6. GPT is the same thing as an AI agent.

False

7. LLMs can transform unstructured text into structured data.

True

8. AI output should always be trusted without validation.

False

---

65. Day 2 Vocabulary

Students should understand these terms.

Large Language Model

A large neural-network model trained to process and generate language.

Parameter

A learned numerical value inside a model.

Token

A text-processing unit used by the model.

Tokenization

Converting text into tokens.

Transformer

The neural-network architecture underlying GPT-style language models.

Attention

A mechanism for modeling relationships among tokens in context.

Context

Information available to the model while generating a response.

Context Window

The limited amount of information a model can consider at once.

Inference

Using a trained model to generate an output.

Temperature

A generation setting, where available, affecting variability in token selection.

Embedding

A numerical vector representation that captures aspects of semantic meaning.

Structured Output

A predictable data format such as JSON.

Deterministic Logic

Explicit rules whose behavior should be predictable.

---

66. Key Day 2 Principles

Have students write down these ten principles.

Principle 1

LLMs process tokens rather than words exactly as humans see them.

Principle 2

GPT generates responses one token at a time.

Principle 3

The next token depends heavily on context.

Principle 4

Context is different from permanent memory.

Principle 5

Model knowledge is different from live external information.

Principle 6

Language models are excellent for fuzzy language tasks.

Principle 7

Traditional code is often better for strict business rules.

Principle 8

Databases should hold exact records; LLMs should interpret language around those records.

Principle 9

Tools extend what an LLM can do.

Principle 10

The most useful AI systems combine multiple components.

---

67. Day 2 Mental Model

Students should leave with this:

```
Prompt
   ↓
Tokens
   ↓
Context
   ↓
Transformer
   ↓
Attention
   ↓
Next-Token Prediction
   ↓
Generated Response
```

But professional AI adds:

```
User
  ↓
Application
  ↓
LLM
  ├── Instructions
  ├── Context
  ├── RAG
  ├── APIs
  ├── Tools
  └── Memory
  ↓
Structured Decision
```

```
 ↓
Business Workflow
```

---

68. Homework Assignment 1 — Token Thinking

Give students five prompts.

For each prompt, identify:

- Instructions
- User data
- Expected output
- Information missing
- External data required

Example:

```
Check whether order #ABC123 has shipped
and write a customer response.
```

Student should answer:

```
Instruction:
Check order and write response.

User Data:
Order #ABC123

Missing:
Actual order status.

External Source:
Order management API.

Expected Output:
Customer-friendly shipping response.
```

Create five examples.

---

69. Homework Assignment 2 — AI or Code?

Classify each task as:

- Best for LLM
- Best for traditional code
- Best as hybrid

Task 1

Analyze sentiment of a customer message.

Expected: LLM

Task 2

Reject any payment over a hard transaction limit.

Expected: Code

Task 3

Analyze a refund request and apply company refund limits.

Expected: Hybrid

Task 4

Summarize customer complaints.

Expected: LLM

Task 5

Calculate 7.25% tax exactly.

Expected: Code / calculator

Task 6

Create a personalized email using CRM data.

Expected: Hybrid

Task 7

Check live inventory and explain availability to a customer.

Expected: Hybrid

---

70. Homework Assignment 3 — Prompt Comparison

Ask the same AI:

Prompt A

```
Analyze this email.
```

Prompt B

```
You are a customer-support analyst.

Analyze the customer email.

Return:
- Summary
- Intent
- Sentiment
- Priority
- Required Action

Email:
[EMAIL]
```

Students record:

```
Prompt A Result:
...

Prompt B Result:
...
```

```
Which was more useful?
Why?
```

This prepares students directly for Day 3.

---

71. Homework Assignment 4 — Business LLM Map

Choose one business.

Examples:

- Ecommerce
- Restaurant
- Hotel
- Real estate
- Recruitment
- Accounting
- Marketing agency

Create a table:

Business Task	LLM Role	External Tool Needed?
Customer email classification	Classify message	No
Check order status	Understand request	Order API
Answer company policy question	Generate answer	RAG
Send follow-up email	Generate content	Email API
Calculate invoice total	Explain result	Calculator/code

Students must create at least **10 rows**.

---

72. Optional Advanced Homework

Design an AI customer-support architecture using:

```
LLM
Database
API
```

```
RAG
Human Approval
```

Example:

```
Customer
    ↓
LLM
    ↓
Intent Detection
    ↓
  ┌─────────────┐
  │Order Issue? │
  └──────┬──────┘
         ↓
     Order API
         ↓
      Live Data
         ↓
        RAG
         ↓
 Company Policies
         ↓
        LLM
         ↓
   Draft Response
         ↓
Sensitive Action?
   ┌─────┴─────┐
 Yes          No
  ↓            ↓
Human       Auto Send
Approval
```

---

73. Instructor Discussion Questions

Use these if time allows.

Question 1

Why shouldn't an LLM replace a database?

Question 2

Why might the same prompt produce two different answers?

Question 3

Why is context important?

Question 4

When should traditional code control a decision instead of AI?

Question 5

Why is JSON useful for AI automation?

Question 6

What information should an LLM retrieve rather than guess?

Question 7

How could an LLM improve a customer support workflow?

---

74. Interview-Style Questions

These are useful even for beginners preparing for AI outsourcing jobs.

What is an LLM?

A large neural-network model trained to process and generate language.

What is a token?

A unit of text processed by a language model.

What is context?

The information available to the model while generating its current response.

What is a context window?

The maximum amount of information that can be considered in a particular model interaction.

What is the difference between GPT and ChatGPT?

GPT refers to model technology/model families, while ChatGPT is an application built around AI models and additional capabilities.

What is the difference between model knowledge and live information?

Model knowledge comes from training and runtime context. Live information usually requires an external source or tool.

Why use an API with GPT?

To allow a software application to send information to the model and connect model outputs with other business systems.

---

75. Day 2 Mini Project

Project: Intelligent Customer Request Analyzer

Input

Customer message.

Required Output

```json
{
  "intent": "",
  "summary": "",
  "sentiment": "",
  "priority": "",
  "entities": {},
  "missing_information": [],
  "requires_external_data": false,
  "recommended_next_action": ""
}
```

---

76. Test Case 1

```
My name is Rachel.

I ordered three monitors last Tuesday.
Order ID is ORD-9938.

They were supposed to arrive yesterday,
but the tracking page hasn't updated.

Can you tell me where they are?
```

Expected concept:


```json
{
  "intent": "order_status",
  "summary": "Customer is asking about three delayed monitors.",
  "sentiment": "concerned",
  "priority": "medium",
  "entities": {
    "customer_name": "Rachel",
    "order_id": "ORD-9938",
    "quantity": 3,
    "product": "monitors"
  },
  "missing_information": [
    "current shipping status"
  ],
  "requires_external_data": true,
  "recommended_next_action": "query order or shipping API"
}
```


---

77. Test Case 2

```
I love the new dashboard.
It is much easier to use than the old version.
```

Possible result:

```json
{
  "intent": "positive_feedback",
  "sentiment": "positive",
  "priority": "low",
  "requires_external_data": false,
  "recommended_next_action": "record customer feedback"
}
```

---

78. Test Case 3

```
Please cancel my subscription immediately.

I was charged $299 this morning even though
I cancelled last week.
```

Possible analysis:

```
Intent:
Billing dispute / cancellation

Sentiment:
Negative

Priority:
High

External data required:
Yes

Required systems:
Billing system
Subscription system

Potential risk:
Financial action
```

This introduces an important concept:

```
AI understands the request.
Business systems verify the facts.
Business rules decide permitted actions.
```

---

79. Instructor Closing Explanation

End Day 2 with:

> An LLM is powerful because it can transform language into useful information and useful information back into language.

But it is not a replacement for:

- Databases
- APIs
- Business rules
- Search systems
- Human judgment
- Security controls

The professional approach is to combine them.

---

80. Day 2 Final Summary

Students should now understand this pipeline:

```
INPUT
 ↓
Tokenization
 ↓
Tokens
 ↓
Context
 ↓
Transformer
 ↓
Attention
 ↓
Next-Token Prediction
 ↓
Generated Tokens
```

```
  ↓
OUTPUT
```

And the more important business architecture:

```
                Customer
                    ↓
                   LLM
              ┌─────┼─────┐
              ↓     ↓     ↓
             RAG   API   Memory
              ↓     ↓
        Knowledge Live Data
              └─────┼─────┘
                    ↓
           Business Rules
                    ↓
                Automation
```

---

81. Day 2 Learning Outcome Checklist

At the end of today's lecture, students should be able to answer:

- [ ] What is an LLM?
- [ ] What does GPT stand for?
- [ ] What is a parameter?
- [ ] What is a token?
- [ ] What is tokenization?
- [ ] What is context?
- [ ] What is a context window?
- [ ] What does attention mean conceptually?
- [ ] How does GPT generate responses?
- [ ] Why might responses differ?
- [ ] What does temperature represent?
- [ ] What is an embedding conceptually?
- [ ] Why shouldn't an LLM replace a database?
- [ ] When should we use deterministic code?
- [ ] Why do AI systems require external tools?
- [ ] How will RAG and APIs extend an LLM?

If students can explain these concepts in simple language, **Day 2 has succeeded.**

---

82. Preview of Day 3

Day 3 — Prompt Engineering Fundamentals

Day 3 moves from:

```
How does GPT work?
```

to:

```
How do we control GPT effectively?
```

Students will learn:

```
Role
+
Goal
+
Context
+
Task
+
Constraints
+
Examples
+
Output Format
=
Better Prompt
```

Day 3 will cover:

- Poor prompts vs good prompts
- Instruction hierarchy
- Role prompting
- Context design
- Constraints
- Few-shot examples
- Output formats
- Prompt templates
- Prompt variables
- Business prompt patterns

- Hallucination reduction
- Prompt testing
- Reusable prompts
- Building a professional customer-support prompt

The Day 3 mini project will be:

Build a reusable Business Customer-Support Prompt Template.